

Lab 3 – Architecture Justification

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Feature Flag Management System

Architecture Selection: I chose Layered Architecture for the Feature Flag Management System.

Architectural Choice: The system is organized into a Presentation Layer, Business Layer, and Data Layer. The Feature Flag Admin UI belongs to the Presentation Layer. The Feature Flag Evaluation Engine, Targeting Rules Engine, and Flag State Synchronization Service belong to the Business Layer. The Flag Configuration Store belongs to the Data Layer.

Reason 1 – Separation of responsibilities: Administrators manage flags through the UI, while evaluation, targeting, and synchronization are handled by dedicated business components. This keeps user-interface changes separate from the core flag-decision logic.

Reason 2 – Maintainable flag configuration: Keeping configuration persistence in a separate data layer gives the evaluation and synchronization components a consistent place to retrieve or update flag definitions, rollout percentages, targeting rules, and environment overrides.

Security Advantage: The admin UI can enforce authenticated, role-based access for flag creation and configuration changes. Separating administrative operations from evaluation requests helps limit configuration changes to authorized users.

Performance Benefit: The Evaluation Engine can use efficient configuration lookups and avoid routing each evaluation through the administrative UI. Keeping evaluation logic focused reduces unnecessary work when applications request flag values.

Key Components: Feature Flag Admin UI; Feature Flag Evaluation Engine; Targeting Rules Engine; Flag Configuration Store; Flag State Synchronization Service.

Key Interfaces: Flag Evaluation API; Targeting Evaluation API; Flag Configuration Query; Flag Configuration API; Flag State Update Event.